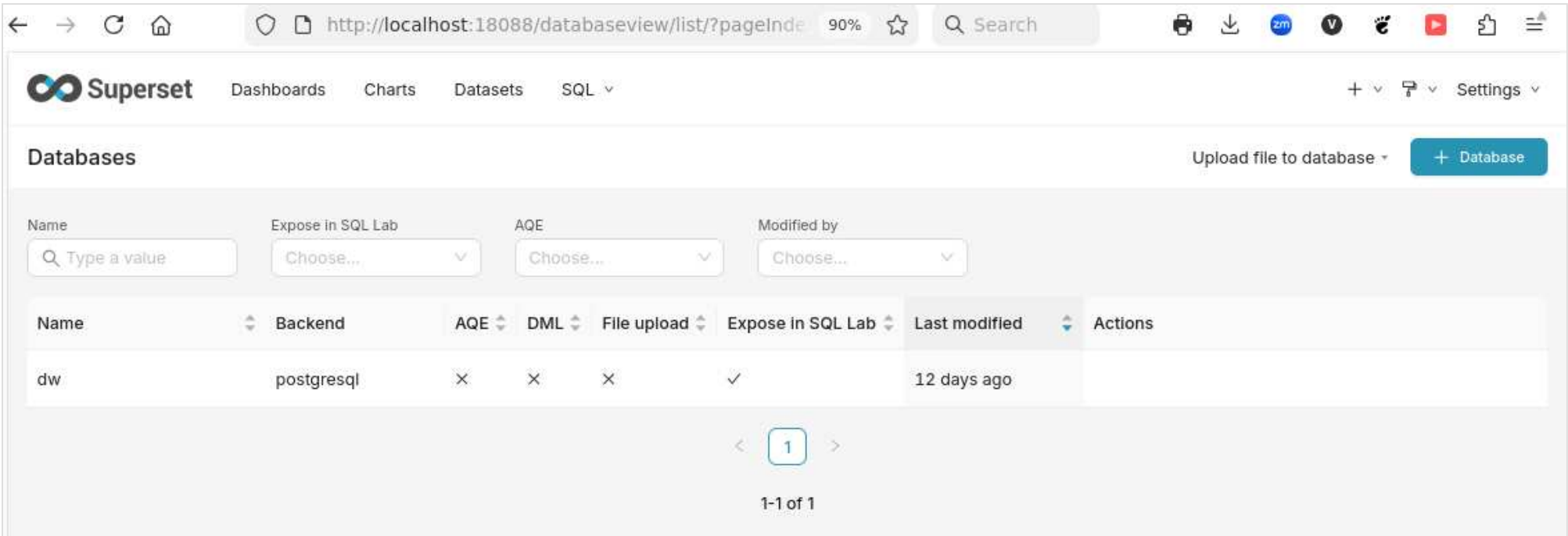


Naloga 6 Overview

This document gives a short visual overview of the Superset environment and the KPI dashboards prepared for the assignment. It is written as a baseline markdown file so it can later be converted into a PDF report.

Superset Environment

The dashboard work is done in Apache Superset. Superset is connected directly to the PostgreSQL data warehouse through the `dw` database connection.



Superset database connection

The KPI charts use virtual datasets in Superset. These datasets are based on SQL files from `na\loga6/sql/`. They keep the warehouse as the source of truth and let Superset apply dashboard filters at query time.

Superset Dashboards Charts **Datasets** SQL + Settings

Datasets [Bulk select](#) [+ Dataset](#)

Name: Type: Database: Schema: Owner: Certified:

Modified by:

Name	Type	Database	Schema	Owners	Last modified	Actions
kpi_kpi_accidents_aqi_state_month_correlation_base ⓘ	Virtual	dw	dw	PA	18 minutes ago	
kpi_kpi_aqi_bad_air_parameter ⓘ	Virtual	dw	dw	PA	19 minutes ago	
kpi_kpi_aqi_bad_air_county ⓘ	Virtual	dw	dw	PA	19 minutes ago	
kpi_kpi_aqi_bad_air_period ⓘ	Virtual	dw	dw	PA	19 minutes ago	
kpi_kpi_accident_count_duration_county ⓘ	Virtual	dw	dw	PA	19 minutes ago	
kpi_kpi_accident_duration_severity_road ⓘ	Virtual	dw	dw	PA	19 minutes ago	
kpi_kpi_accident_duration_weather ⓘ	Virtual	dw	dw	PA	19 minutes ago	
kpi_kpi_accident_duration_period ⓘ	Virtual	dw	dw	PA	19 minutes ago	
kpi_kpi_accident_count_county ⓘ	Virtual	dw	dw	PA	19 minutes ago	
kpi_kpi_accident_count_severity_time ⓘ	Virtual	dw	dw	PA	19 minutes ago	
kpi_kpi_accident_count_hour_dow ⓘ	Virtual	dw	dw	PA	19 minutes ago	
kpi_kpi_accident_count_period ⓘ	Virtual	dw	dw	PA	19 minutes ago	
kpi_kpi_accidents_aqi_state_correlation ⓘ	Virtual	dw	dw	PA	a day ago	
kpi_kpi_accidents_aqi_correlation_summary ⓘ	Virtual	dw	dw	PA	a day ago	
kpi_kpi_accidents_vs_aqi_monthly ⓘ	Virtual	dw	dw	PA	a day ago	
dw_dim_time_preview ⓘ	Virtual	dw	dw	PA	12 days ago	

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Superset KPI datasets

The chart list shows the generated KPI charts. It includes table-preview charts for SQL inspection and graphical charts used on the dashboards.

Superset Dashboards Charts Datasets SQL

Charts

Name: Type: Dataset: Owner: Dashboard:

Favorite: Certified: Modified by:

Name	Type	Dataset	On dashboards	Owners
KPI - KPI2 Top Counties by Accident Count	Bar Chart	kpi_kpi_accident_count_duration_county	KPI 2 - Accident Duration Dashboard	PA
KPI - KPI2 Duration by Severity and Road SI...	Bar Chart	kpi_kpi_accident_duration_severity_road	KPI 2 - Accident Duration Dashboard	PA
KPI - KPI2 Duration by Weather	Bar Chart	kpi_kpi_accident_duration_weather	KPI 2 - Accident Duration Dashboard	PA
KPI - KPI2 Duration Trend	Line Chart	kpi_kpi_accident_duration_period	KPI 2 - Accident Duration Dashboard	PA
KPI - KPI2 Median Accident Duration	Big Number with Trendline	kpi_kpi_accident_duration_period	KPI 2 - Accident Duration Dashboard	PA
KPI - State AQI Accident Correlation Ranking	Bar Chart	kpi_kpi_accidents_aqi_state_month_correlati...	Dashboard 4 - Accidents and Air Quality Re...	PA
KPI - AQI Severity Correlation	Big Number	kpi_kpi_accidents_aqi_state_month_correlati...	Dashboard 4 - Accidents and Air Quality Re...	PA
KPI - AQI Duration Correlation	Big Number	kpi_kpi_accidents_aqi_state_month_correlati...	Dashboard 4 - Accidents and Air Quality Re...	PA
KPI - AQI Accident Count Correlation	Big Number	kpi_kpi_accidents_aqi_state_month_correlati...	Dashboard 4 - Accidents and Air Quality Re...	PA
KPI - KPI3 Bad-Air by Defining Parameter	Pie Chart	kpi_kpi_aqi_bad_air_parameter	KPI 3 - Air Quality Dashboard	PA
KPI - KPI3 Counties by Bad-Air Share	Bar Chart	kpi_kpi_aqi_bad_air_county	KPI 3 - Air Quality Dashboard	PA
KPI - KPI3 Bad-Air Share Trend	Line Chart	kpi_kpi_aqi_bad_air_period	KPI 3 - Air Quality Dashboard	PA
KPI - KPI3 Bad-Air AQI Share	Big Number with Trendline	kpi_kpi_aqi_bad_air_period	KPI 3 - Air Quality Dashboard	PA
KPI - KPI2 Count vs Duration by County	Scatter Plot	kpi_kpi_accident_count_duration_county		PA
KPI - KPI2 Average Accident Duration	Big Number with Trendline	kpi_kpi_accident_duration_period		PA
KPI SQL Preview - 11_kpi_aqi_bad_air_para...	Table	kpi_kpi_aqi_bad_air_parameter		PA
KPI SQL Preview - 10_kpi_aqi_bad_air_county	Table	kpi_kpi_aqi_bad_air_county		PA
KPI SQL Preview - 09_kpi_aqi_bad_air_period	Table	kpi_kpi_aqi_bad_air_period		PA

Superset KPI charts

The dashboard list shows the published dashboards used for this assignment:

- KPI 1 - Accident Frequency Dashboard
- KPI 2 - Accident Duration Dashboard
- KPI 3 - Air Quality Dashboard
- Dashboard 4 - Accidents and Air Quality Relationship

← → ↻ 🏠 <http://localhost:18088/dashboard/list/?pageIndex=1> 90% ☆ 🔍 Search 🖨️ ⬇️ zm V 🦉 📺 ⌵ ⌵ ⌵

Superset [Dashboards](#) [Charts](#) [Datasets](#) [SQL](#) + ⌵ [Settings](#) ⌵

Dashboards

⬇️ [Bulk select](#) [+ Dashboard](#)

Name: Status: Owner: Favorite: Certified:

Modified by:

Name	Status	Owners	Last modified	Actions
☆ KPI 2 - Accident Duration Dashboard	☒ Published	PA	15 minutes ago	
☆ KPI 3 - Air Quality Dashboard	☒ Published	PA	an hour ago	
☆ Dashboard 4 - Accidents and Air Quality Relationship	☒ Published	PA	5 hours ago	
☆ KPI 1 - Accident Frequency Dashboard	☒ Published	PA	6 hours ago	
☆ DW Superset Proof Dashboard	☒ Published	PA	12 days ago	

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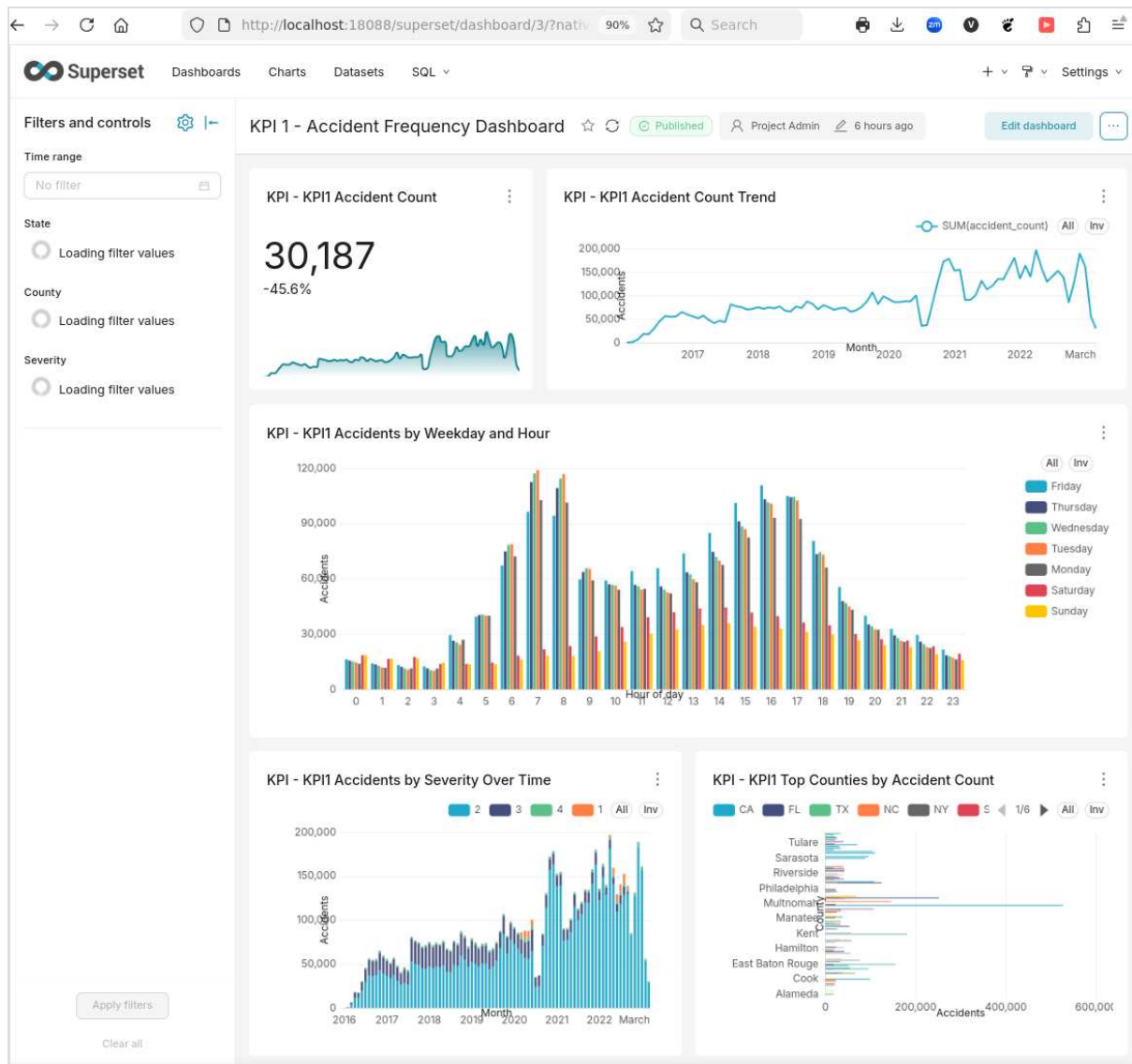
Superset dashboards

KPI 1: Accident Frequency

The first dashboard focuses on accident counts. It answers when and where accidents are most frequent, and how accident volume changes by severity.

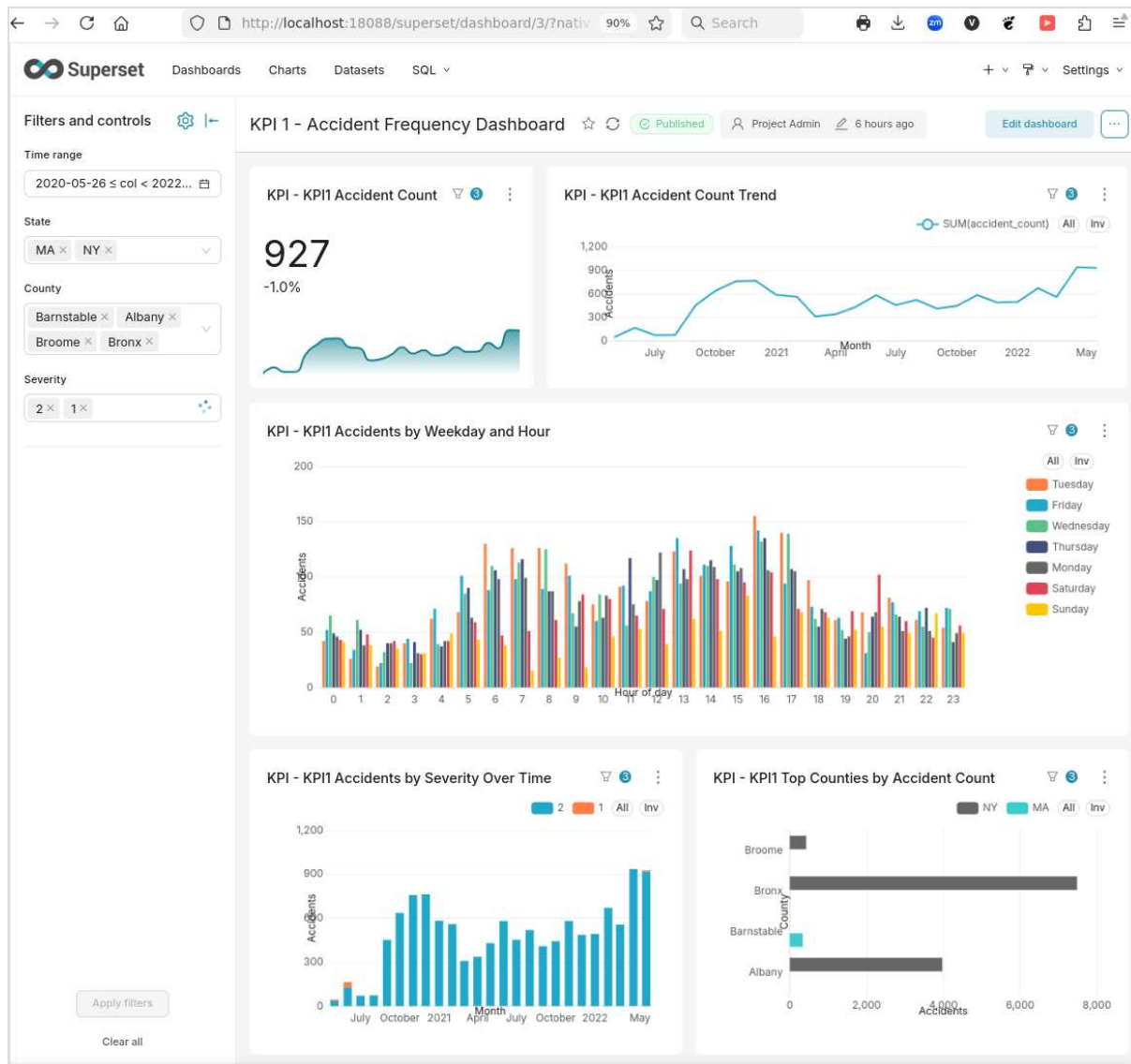
The unfiltered view shows the full accident dataset. The dashboard includes:

- a KPI card with monthly accident count and trend;
- a monthly accident-count trend chart;
- weekday and hour accident patterns;
- accident counts split by severity over time;
- top counties by accident count.



KPI 1 accident frequency without filters

The filtered view shows the same charts after applying a time range, selected states, selected counties, and selected severity levels. The charts recalculate from the filtered data, so the large number, trend, weekday/hour chart, severity chart, and county ranking all describe the selected context.



KPI 1 accident frequency with filters

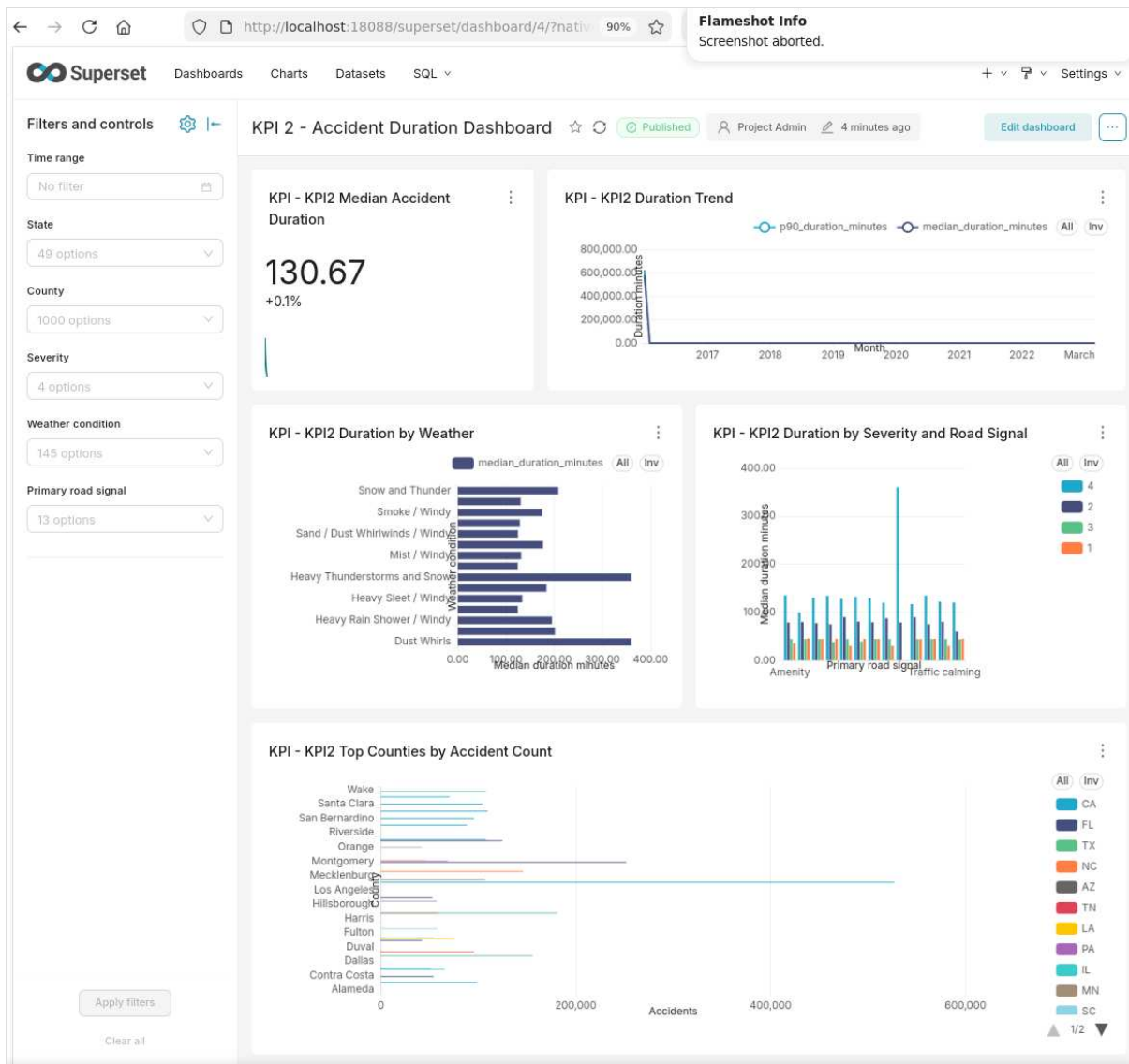
This dashboard is useful for finding accident concentration by time and geography. It also shows how filters narrow the analysis from national-level data to a smaller local context.

KPI 2: Accident Duration

The second dashboard focuses on accident duration. During inspection, average duration was not readable enough because a few very long events dominated the chart scale. The dashboard was therefore changed to use median duration and p90 duration as the main visible duration measures. The outliers remain in the data, but they no longer hide the normal pattern.

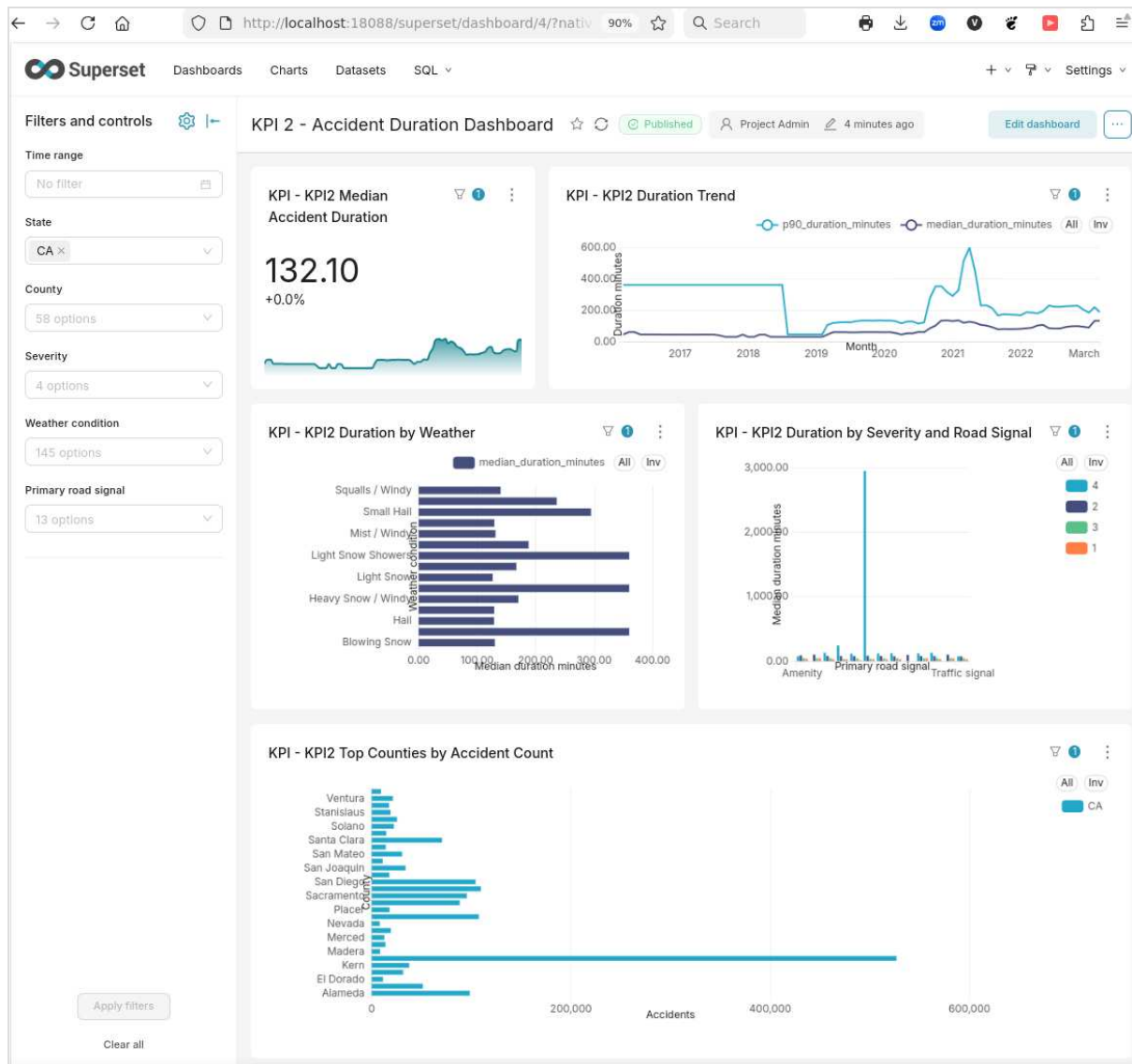
The unfiltered view includes:

- a KPI card for median accident duration;
- a monthly trend with median duration and p90 duration;
- median duration by weather condition;
- median duration by severity and primary road signal;
- top counties by accident count.



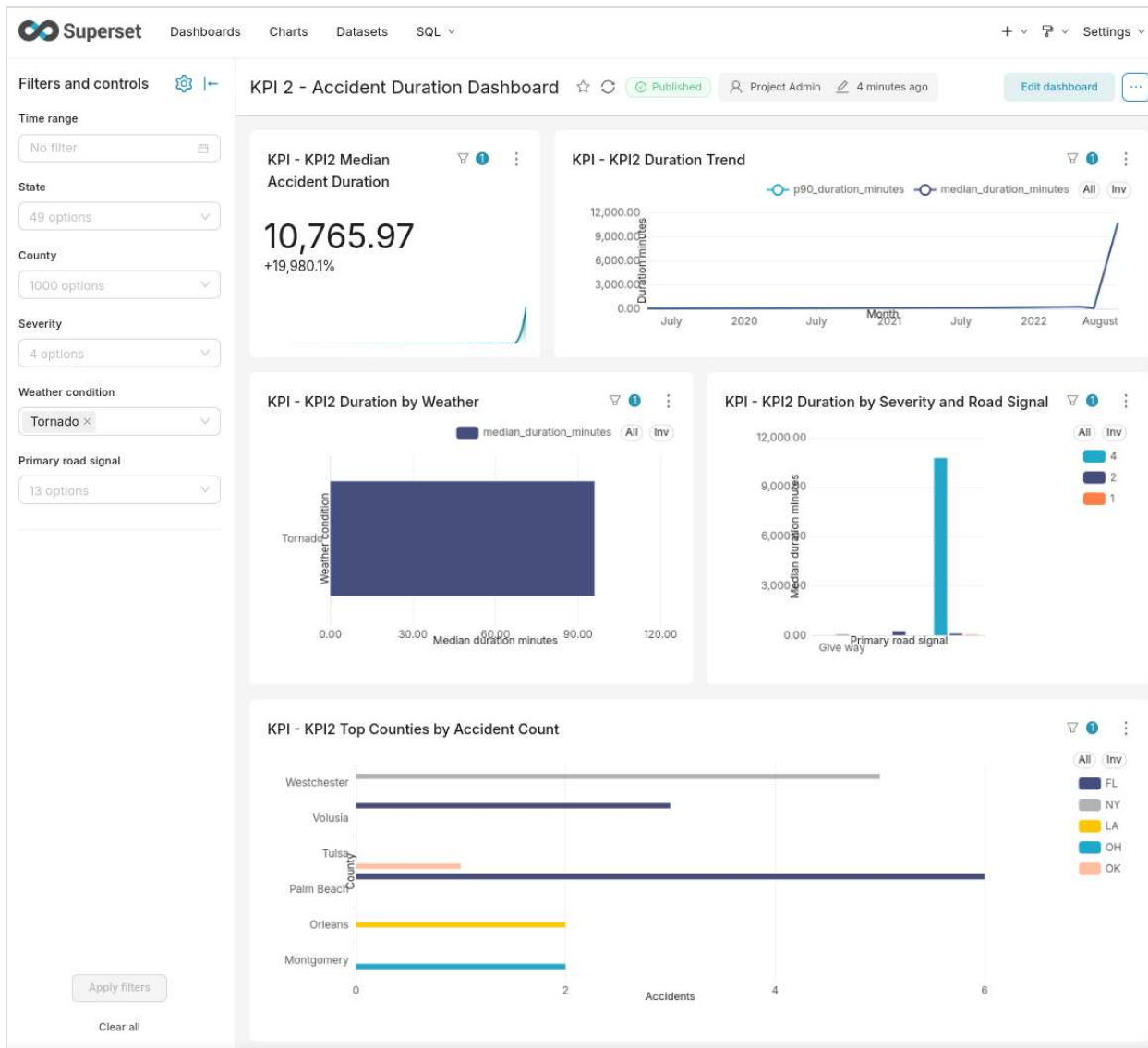
KPI 2 accident duration without filters

The California-filtered view shows how the same dashboard changes when a state filter is applied. The charts now describe California only, and the county ranking becomes a California county ranking.



KPI 2 accident duration filtered to California

The weather-filtered view shows a focused case for the **Tornado** weather condition. This is a small and unusual subset of the data. It demonstrates that the dashboard can drill into rare conditions without removing outliers from the source data.



KPI 2 accident duration filtered to tornado weather

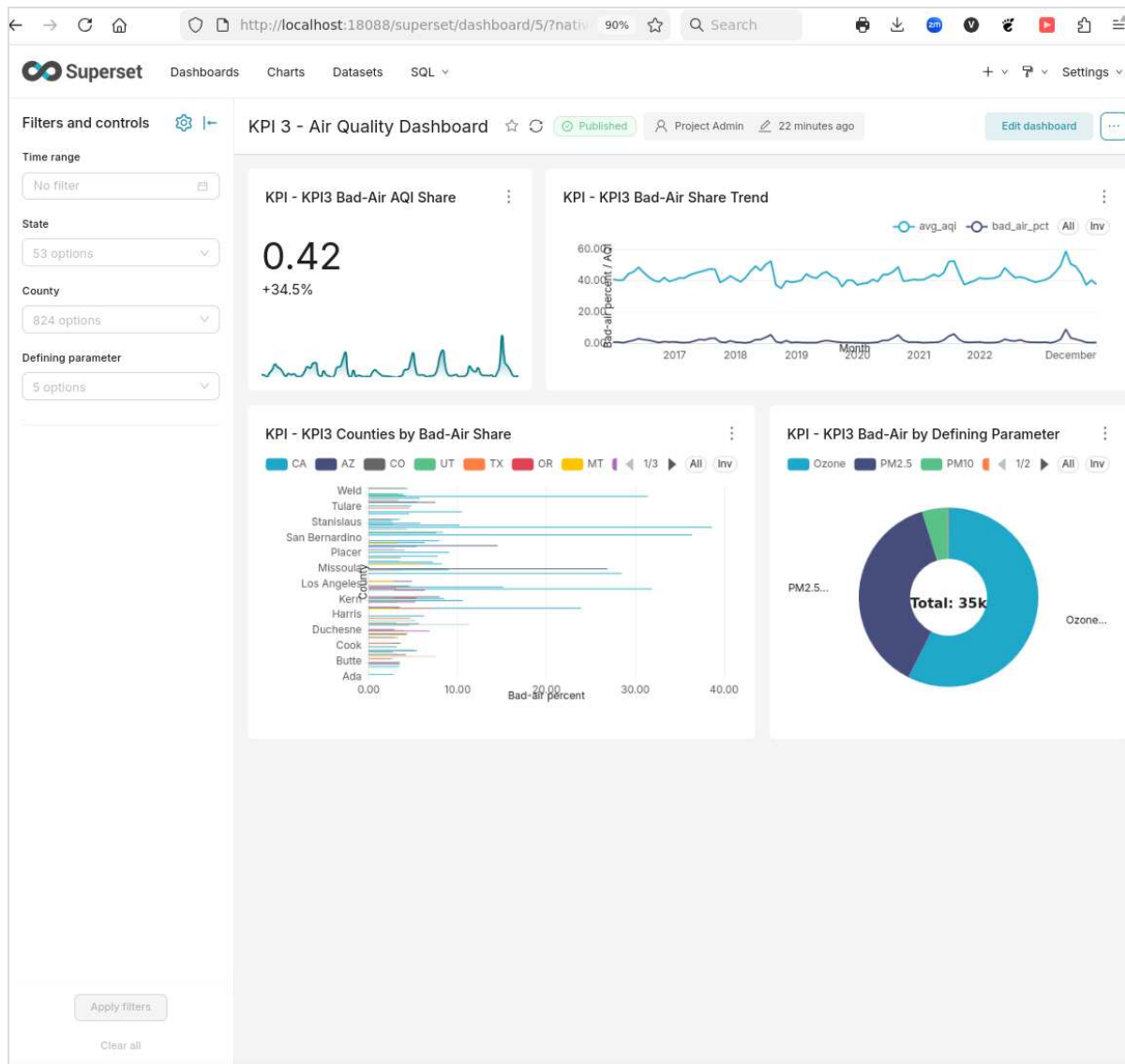
This dashboard is useful for comparing typical duration across weather, severity, road signal, and geography. It also gives accident volume context through the top-county chart.

KPI 3: Air Quality

The third dashboard focuses on bad-air AQI observations. Bad air is defined as `AQI > 100` , following the assignment proposal.

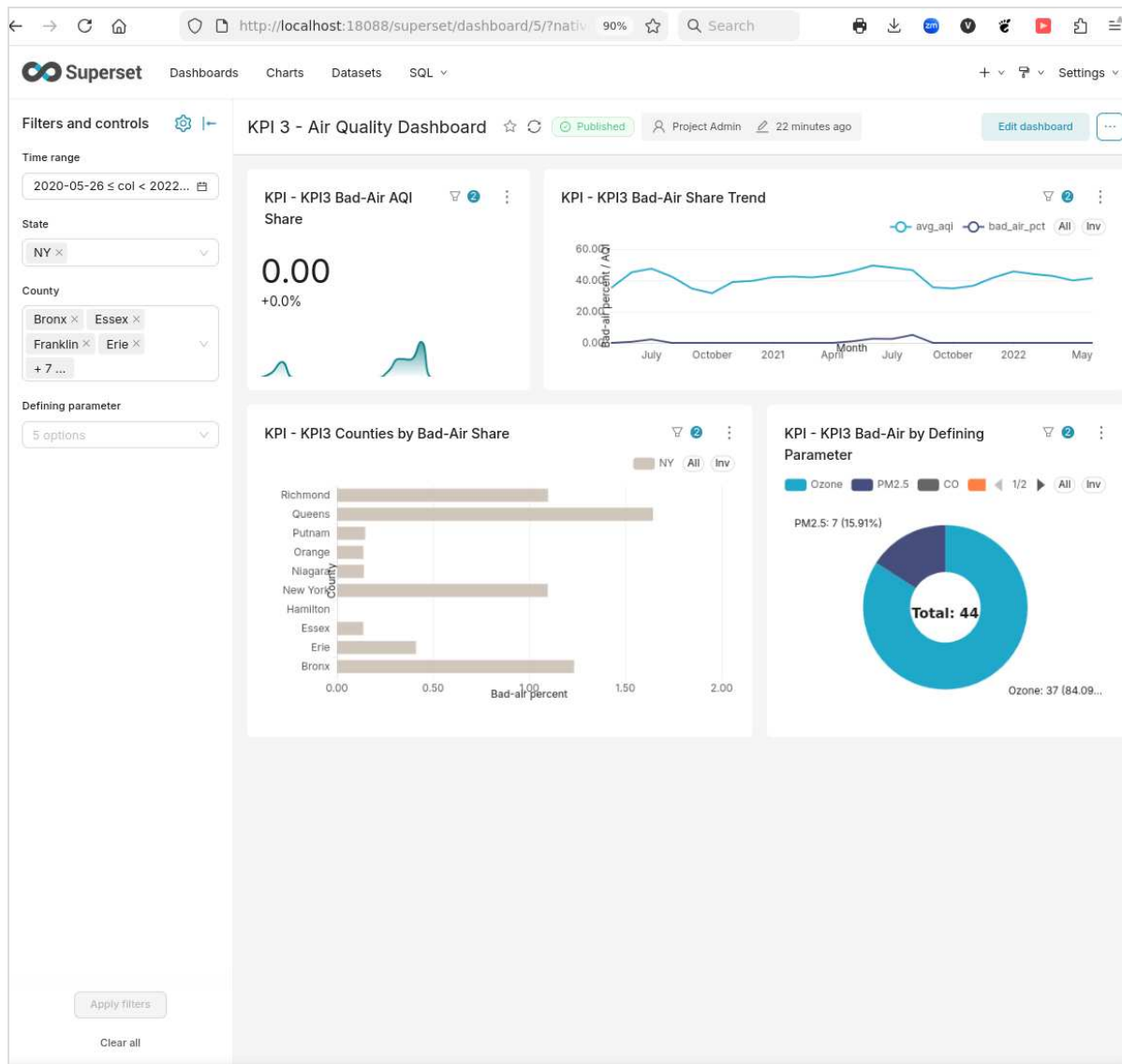
The unfiltered view includes:

- a KPI card for bad-air AQI share;
- a trend chart comparing average AQI and bad-air share;
- counties ranked by bad-air share;
- a donut chart showing which defining parameter contributes to bad-air observations.



KPI 3 air quality without filters

The filtered view shows the dashboard after applying a time range, a state filter, and a county selection. The county ranking and defining-parameter chart now describe only the selected New York counties and selected period.



KPI 3 air quality with filters

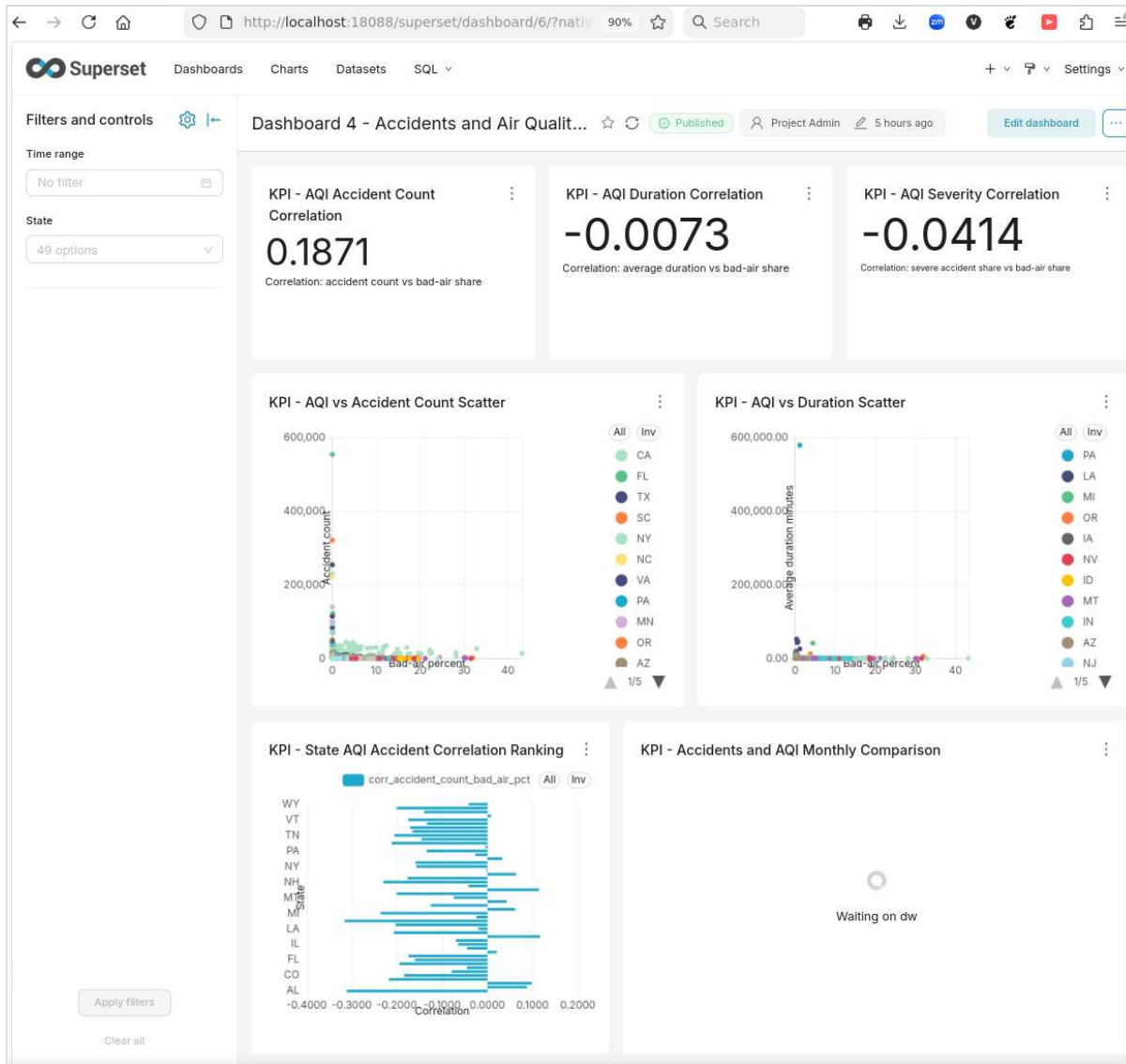
This dashboard is useful for seeing where bad-air observations occur and which pollutant parameter is most often responsible for those observations.

Dashboard 4: Accidents and Air Quality Relationship

The final dashboard combines accident and air-quality measures. It works at state/month grain and is meant for relationship analysis, not causal proof.

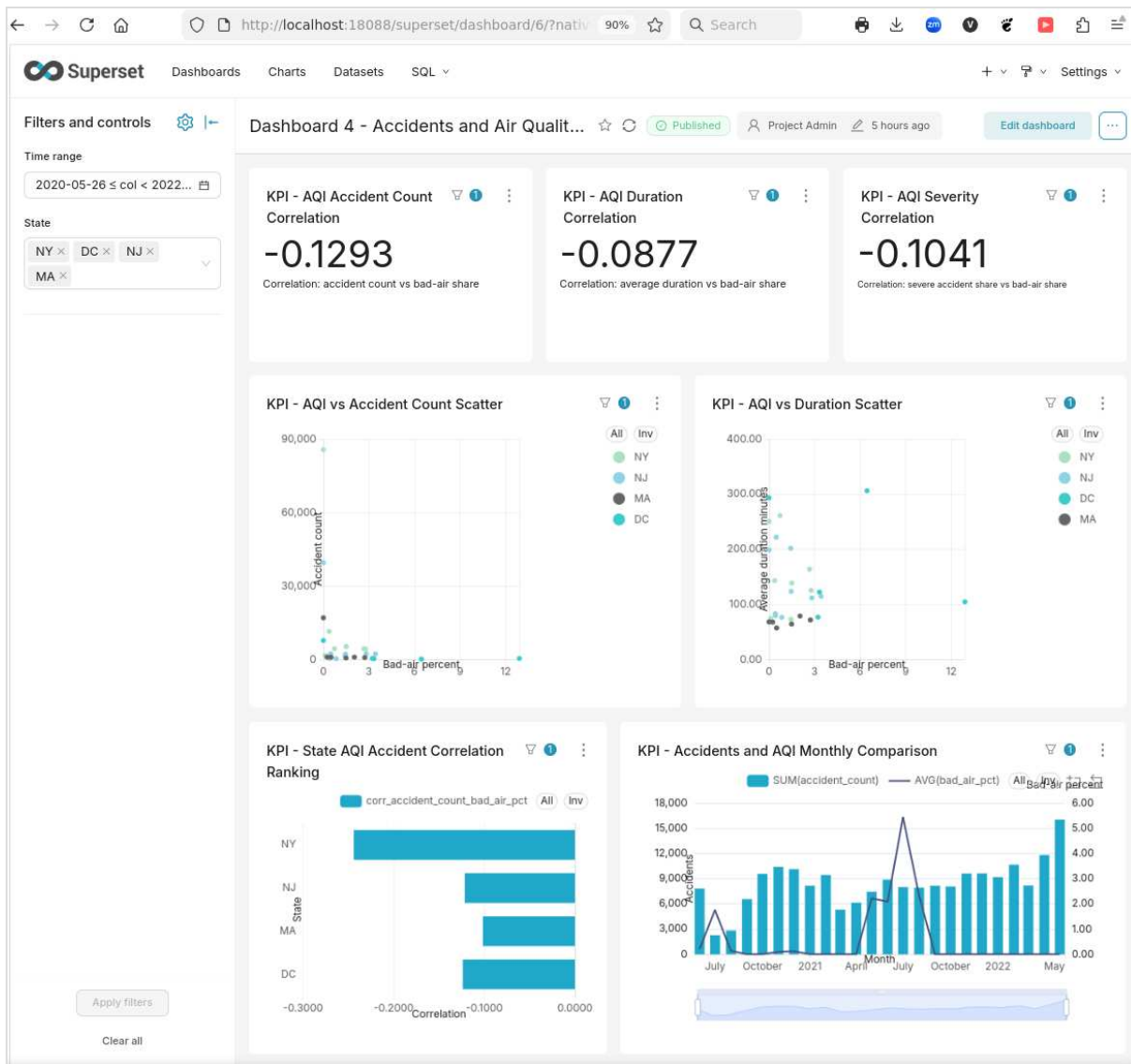
The unfiltered view shows the overall relationship between accident measures and bad-air share:

- correlation between accident count and bad-air share;
- correlation between average accident duration and bad-air share;
- correlation between severe accident share and bad-air share;
- scatter plots for accident count and duration against bad-air share;
- a state correlation ranking;
- a monthly comparison of accident count and bad-air percent.



Combined accidents and air quality without filters

The filtered view applies a time range and selected states. The correlation cards and charts recompute from the selected state/month observations. In this example, the selected states show weak negative correlations.



Combined accidents and air quality with filters

The combined dashboard should be read carefully. A positive correlation means two measures tend to rise together. A negative correlation means one tends to rise while the other falls. Values near zero mean there is no strong linear relationship in the selected data. These charts describe association only;

they do not prove that air quality causes accident changes.

Summary

The Superset setup now contains a complete dashboard set for the assignment:

- KPI 1 shows accident frequency.
- KPI 2 shows accident duration using median and p90 duration.
- KPI 3 shows bad-air AQI share.
- Dashboard 4 compares accident measures with air-quality measures.

All dashboards use warehouse-backed virtual datasets. Dashboard filters and cross-filtering are configured for the visible chart dimensions. Outliers are kept in the data, and the visual choices are adjusted where needed so the dashboards remain readable.